Exp No : 10 (c) DESIGNING AN INTER-NETWORK
28/09/26 USING SWITCH, ROUTER & ETERNET.

AIM:

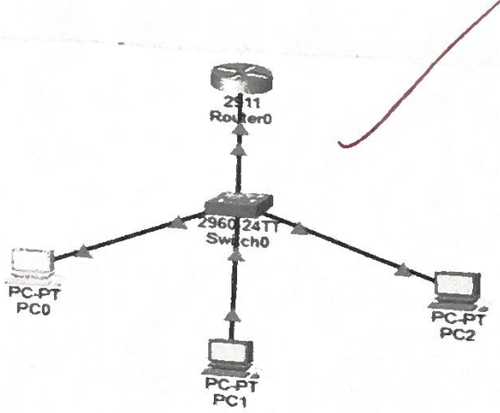
To design and configure an internetwork using a switch, router and ethernet cables in cisco packet server.

THEORY:

A switch connects multiple devices within a LAN, while a router connects and manages communication between networks. Ethernet cables are used to establish wired connections between the devices.

PROCEDURE:

1. Place 1 router, 1 switch and 3 PCs.
2. Connect them using copper straight-through cables.
3. Configure router G1/0 as 192.168.30.1/24.
4. Assign PC0, PC1 & PC2.
5. set the default gateways as 192.168.30.1.
6. Use ping to verify connectivity between the PCs.



```
C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=26ms TTL=125
Reply from 20.0.0.2: bytes=32 time=18ms TTL=125
Reply from 20.0.0.2: bytes=32 time=14ms TTL=125
Reply from 20.0.0.2: bytes=32 time=14ms TTL=125

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 26ms, Average = 18ms

C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    1 ms      0 ms      1 ms      192.168.1.250
  3   10 ms      1 ms      1 ms      192.168.1.245
  4    5 ms      0 ms      3 ms      20.0.0.2

Trace complete.
```

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

RESULT :

Thus, the internetwork was sucessfully configured using a switch, router and ethernet cables, and connectivity was verified using ping command.